

SUNDRAM PATHAK

Hamirpur, Uttar Pradesh, India

+91-6307342675

pathaksundram82@gmail.com

LinkedIn

GitHub

LeetCode

Website

EDUCATION

Kamla Nehru Institute Of Technology, Sultanpur
B.Tech - Electronics Engineering - CGPA : 8.25

Expected May 2026
Sultanpur, Uttar Pradesh

PROJECTS

DevGraph | Next.js, Node.js, Express.js, Supabase, PostgreSQL, Tailwind CSS, Docker, Vercel, Render

- Architected a full-stack developer knowledge management platform supporting 7 structured note categories (bug-fix, snippet, architecture, command, config, learning, other) for organized capture of debugging artifacts and code snippets.
- Implemented a 3-layer hybrid search pipeline combining FlexSearch prefix indexing, PostgreSQL full-text search (TSVECTOR), and substring fallback queries, enabling fast retrieval of developer notes from a unified index.
- Developed an interactive knowledge graph visualization generating node relationships from shared tags and similarity heuristics, enabling exploration of connected technical notes through dynamic graph rendering.
- Secured backend APIs using 5+ security controls including Supabase Auth, Row Level Security (RLS) policies, rate limiting, request validation, and XSS sanitization to enforce user-scoped access to private notes.
- Deployed the platform across 2 cloud environments with a Next.js frontend on Vercel and a containerized Express API on Render using Docker, enabling scalable note storage and search across thousands of entries.

Bank Transaction System | React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Nodemailer, Render

- Engineered a production-ready full-stack banking platform with JWT authentication, bcrypt hashing, token blacklisting, and httpOnly cookie security — mitigating 20+ API abuse vectors per minute via route-level rate limiting.
- Established a double-entry ledger accounting model using MongoDB ACID transactions and atomic session-based fund transfers, ensuring 100% financial consistency with immutable DEBIT/CREDIT entries across all operations.
- Architected RESTful APIs across 5 resource domains (auth, accounts, transactions, profile, admin) with idempotency key support, paginated transaction history, and server-side CSV streaming for 1,000+ record exports.
- Built a professional dark-themed React SPA with a persistent sidebar, double-column ledger table (separate debit/credit columns with running sequence numbers), real-time filter strip, and a ledger summary bar — mirroring real financial institution UI conventions.
- Deployed a request-level audit logging system using non-blocking res.on('finish') middleware capturing userId, IP, route, method, and response time into a dedicated audit_logs collection for full operational traceability.

AlgoVista | React.js, JS, Tailwind CSS, Framer Motion, shadcn/ui, Radix UI, React Router, Vercel

- Engineered a frontend-only algorithm learning platform with 3 core user flows, 4 client-side routes, and a zero-backend deployment model, enabling seamless access to visual simulations and reference content without server-side dependencies.
- Built 6 visualization systems for sorting, searching, graphs, trees, linked lists, and stack/queue operations, with playback controls for stepping, reset, randomization, and speed tuning.
- Structured a local encyclopedia of 200+ algorithms with difficulty levels, complexity analysis, walkthroughs, and multi-language implementations across 4 programming languages.
- Developed a reusable UI architecture using 50+ component abstractions with responsive design, accessible interaction patterns, and motion-driven onboarding to improve learning-focused usability.
- Launched the application on Vercel and prepared it for public release with production-ready documentation, licensing, branding assets, and secret-safe repository hygiene.

LEADERSHIP AND RESPONSIBILITY

Career Development Cell, KNIT
Training and Placement Representative

Feb 2024 – Present

Spearheaded LinkedIn outreach initiatives and organized internship and placement drives by partnering with 15+ companies, supporting 150+ students, and optimizing recruitment workflows—boosting placement success rate by 25%.

SKILLS

Languages: Python, C, C++, JavaScript, HTML5, CSS3, SQL

Developer Tools: VS Code, Antigravity, Cursor, WindSurf, Claude, Git, GitHub, Render, Vercel, Codex

Technologies/Frameworks: Tailwind CSS, ReactJS, NextJS, NodeJS, ExpressJS

Databases: MongoDB, MySQL, Supabase

Coursework: DSA, OOPs, OS, CN, DBMS

Professional Skills: Cross-functional collaboration, Technical communication, Analytical thinking, Structured debugging

ACHIEVEMENTS

- 3* Rated** on CodeChef, demonstrating strong algorithmic problem-solving and competitive programming proficiency.
- Guardian Rating** on LeetCode, reflecting consistent performance in advanced data structures and algorithms challenges.
- Solved **500+ DSA problems** across platforms including LeetCode, CodeChef, and GeeksforGeeks.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Oct 2025 - Oct 2027